

A Preference-Conditioned Multi-Objective Deep Reinforcement Learning for the Rate-Energy-Fairness Tradeoff in Multiuser SWIPT Systems

Jeongwon Kim

*School of Electrical Engineering
Korea University
Seoul, South Korea
jeongwonkim@korea.ac.kr*

Sangmin Kim

*School of Electrical Engineering
Korea University
Seoul, South Korea
smgeem@korea.ac.kr*

Zhilin Fu

*Samsung Electronics Co., Ltd.
Suwon, South Korea
celynnfu@korea.ac.kr*

Jihwan Moon

*Dept. of Mobile Convergence Engineering
Hanbat National University
Daejeon, South Korea
anschino@staff.hanbat.ac.kr*

Sangwon Hwang

*Dept. of Computer and AI Engineering
Pukyong National University
Busan, South Korea
s.won.hwang@pknu.ac.kr*

Inkyu Lee

*School of Electrical Engineering
Korea University
Seoul, South Korea
inkyu@korea.ac.kr*

Abstract—In this paper, we study joint rate, harvested energy, and user fairness control in a multiuser simultaneous wireless information and power transfer (SWIPT) system with power-splitting receivers. We formulate the problem as a battery-aware multi-objective Markov decision process (MDP) under transmit power and battery dynamics, in which sum rate, harvested energy, and user fairness are optimized simultaneously via a vector reward. We solve it using a preference-conditioned multi-objective deep deterministic policy gradient (MO-DDPG) approach that learns a single policy over the preference space. Simulation results show that the proposed single policy network achieves performance comparable to preference-specialized baseline policies while reducing training and storage overhead, and enables real-time priority adaptation during deployment without policy retraining.

Index Terms—SWIPT, rate-energy-fairness tradeoff, multi-objective reinforcement learning, DDPG, power splitting, fairness.

I. INTRODUCTION

Radio-frequency (RF) signals enable simultaneous wireless information and power transfer (SWIPT) for low-power Internet-of-Things devices [1]–[3]. Practical SWIPT receivers employ a power-splitting (PS) architecture that divides received power between information decoding (ID) and energy harvesting (EH) [4]. In multiuser systems, joint power allocation and PS control induce a fundamental three-objective

tradeoff among sum rate, harvested energy, and user fairness, where allocating power to channel-favorable users maximizes sum rate while serving channel-impaired users ensures fairness.

Conventional optimization approaches determine a single operating point along this tradeoff boundary for a fixed preference weight [4], [5], requiring full re-optimization whenever operational priorities change. Under time-varying channels and EH dynamics, deep reinforcement learning (DRL) algorithms such as deep deterministic policy gradient (DDPG) [6] effectively address sequential decision-making in wireless networks [7]–[9]. However, standard DRL agents optimize a single scalar reward, necessitating N separately trained policies to cover the tradeoff boundary. Multi-objective RL instead learns policies that generalize across preferences [10]. In this paper, we apply preference-conditioned MO-DDPG to multiuser SWIPT to control sum rate, harvested energy, and user fairness within a single policy network. The main contributions of this work are summarized as follows:

- We formulate multiuser PS-SWIPT with finite batteries as a multi-objective Markov decision process (MDP) under transmit power and battery dynamics, capturing the three-way rate-energy-fairness tradeoff.
- We propose a preference-conditioned MO-DDPG framework with a vector critic that uses a single network to approximate the rate-energy-fairness tradeoff region, while achieving performance comparable to preference-specialized baseline policies at lower training and storage overhead.
- We demonstrate real-time priority adaptation during on-

This work was supported in part by the National Research Foundation of Korea (NRF) grant funded by the Korea government (MSIT) under Grant RS-2022-NR070834 and Grant RS-2026-25599154, and in part by the IITP(Institute of Information Communications Technology Planning Evaluation)-ITRC(Information Technology Research Center) grant funded by the Korea government(Ministry of Science and ICT)under Grant IITP-2026-RS-2024-00437886 (33%).

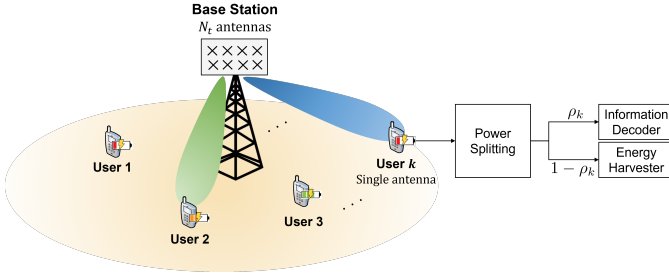


Fig. 1. System model of a multiuser MISO power-splitting SWIPT network with finite battery energy storage.

line deployment without network retraining.

II. SYSTEM MODEL AND PROBLEM FORMULATION

We consider a downlink SWIPT system, as depicted in Fig. 1, in which a base station (BS) with N_t antennas serves K single-antenna power-splitting users, indexed by $k \in \mathcal{K} \triangleq \{1, \dots, K\}$. The signal model follows the multiuser multiple-input single-output (MISO) power-splitting SWIPT model of [4]. Time is slotted with slot duration τ , and the channel $\mathbf{h}_k[t] \in \mathbb{C}^{N_t}$ is constant within a slot and independent across slots, with $\mathbf{h}_k[t] = \sqrt{\beta_k} \tilde{\mathbf{h}}_k[t]$, $\tilde{\mathbf{h}}_k[t] \sim \mathcal{CN}(\mathbf{0}, \mathbf{I})$, and large-scale gain β_k .

The BS uses maximum-ratio transmission (MRT), $\mathbf{v}_k[t] = \mathbf{h}_k[t] / \|\mathbf{h}_k[t]\|$, with transmit power $p_k[t] \geq 0$, and transmits $\mathbf{x}[t] = \sum_k \sqrt{p_k[t]} \mathbf{v}_k[t] s_k[t]$. Defining the effective gain $g_{k,j}[t] \triangleq |\mathbf{h}_k[t]^H \mathbf{v}_j[t]|^2$, user k employs a PS ratio $\rho_k[t] \in [0, 1]$ to adjust the proportion of received power allocated to ID and EH. Considering both the antenna noise $z_{a,k}[t] \sim \mathcal{CN}(0, \sigma_a^2)$ and the noise added at the ID receiver $z_{c,k}[t] \sim \mathcal{CN}(0, \sigma_c^2)$, the signal-to-interference-plus-noise ratio (SINR) and rate are given by

$$\gamma_k[t] = \frac{\rho_k[t] p_k[t] g_{k,k}[t]}{\sum_{j \neq k} \rho_k[t] p_j[t] g_{k,j}[t] + \rho_k[t] \sigma_a^2 + \sigma_c^2}, \quad (1)$$

$$R_k[t] = \log_2(1 + \gamma_k[t]). \quad (2)$$

With MRT beamforming, the active control variables at slot t are the transmit powers $\{p_k[t]\}$ and power-splitting ratios $\{\rho_k[t]\}$. Following the linear EH model [4], the harvested energy at user k is given by

$$E_k[t] = \eta (1 - \rho_k[t]) \tau \left(\sum_{j \in \mathcal{K}} p_j[t] g_{k,j}[t] + \sigma_a^2 \right), \quad (3)$$

where $\eta \in (0, 1]$ denotes the RF-to-DC energy conversion efficiency. Each user stores energy in a battery of capacity $B_{\max}$ and, when active, consumes E_{op} per slot. A user can decode only if its battery can pay this cost. With the operation indicator $o_k[t] = \mathbf{1}\{b_k[t] \geq E_{\text{op}}\}$, the effective rate is given by $o_k[t] R_k[t]$. Consequently, a depleted user enters an outage state with zero effective rate, and the battery state evolves as

$$b_k[t+1] = \min\{B_{\max}, [b_k[t] - o_k[t] E_{\text{op}} + E_k[t]]^+\}, \quad (4)$$

where $[\cdot]^+ = \max\{\cdot, 0\}$, and EH is passive in that it does not draw power from the battery. The battery dynamics couple de-

cisions across slots, making this a sequential decision problem that warrants an MDP formulation.

The BS controls the power allocation $\mathbf{p}[t] = (p_1[t], \dots, p_K[t])$ and the PS ratios $\boldsymbol{\rho}[t] = (\rho_1[t], \dots, \rho_K[t])$ to jointly optimize three long-term objectives: sum rate, harvested energy, and user fairness (defined as the max-min effective rate). Using the operation indicator $o_k[t]$ defined in (4), the per-slot utility of each objective is given by

$$u_R[t] = \sum_{k \in \mathcal{K}} o_k[t] R_k[t], \quad (5)$$

$$u_E[t] = \sum_{k \in \mathcal{K}} E_k[t], \quad (6)$$

$$u_F[t] = \min_{k \in \mathcal{K}} o_k[t] R_k[t], \quad (7)$$

and each objective is formulated as its long-term average utility over time, given by

$$f_i = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T u_i[t], \quad i \in \{R, E, F\}. \quad (8)$$

To effectively characterize the Pareto-optimal trade-off region among sum rate, harvested energy, and user fairness, the joint power allocation and power-splitting design is formulated as the following multi-objective optimization problem:

$$\max_{\{\mathbf{p}[t], \boldsymbol{\rho}[t]\}} (f_R, f_E, f_F) \quad (9a)$$

$$\text{s.t. (C1)} \quad \sum_{k \in \mathcal{K}} p_k[t] \leq P_{\max}, \quad \forall t, \quad (9b)$$

$$\text{(C2)} \quad 0 \leq \rho_k[t] \leq 1, \quad \forall k, t, \quad (9c)$$

$$\text{(C3)} \quad b_k[t] \geq 0, \quad \forall k, t, \quad (9d)$$

where (C1) is the transmit power constraint, (C2) is the PS ratio constraint, and (C3) is the battery energy constraint. To approximate the three-objective tradeoff region while satisfying constraints (C1)–(C3), we design a preference-conditioned DRL framework using a single trained policy network.

III. PREFERENCE-CONDITIONED MO-DDPG

We formulate problem (9) as a multi-objective MDP $(\mathcal{S}, \mathcal{A}, \mathbf{r}, \gamma)$ augmented with a preference space $\Omega = \{\boldsymbol{\omega} \in \mathbb{R}_{\geq 0}^3 : \omega_R + \omega_E + \omega_F = 1\}$, where $\boldsymbol{\omega} = (\omega_R, \omega_E, \omega_F)$ weighs sum rate, harvested energy, and user fairness. Constraints (C1)–(C3) are satisfied by construction through action space mappings and state transition dynamics.

A. State, Action, and Reward

1) *State*: At the start of slot t , the BS observes log-normalized effective channel gains and normalized battery levels. Specifically, defining $\bar{g}_{k,j}[t] = \log(1 + g_{k,j}[t]/\sigma_a^2)$, the state is

$$\mathbf{s}[t] = (\{\bar{g}_{k,j}[t]\}_{k,j \in \mathcal{K}}, \{b_k[t]/B_{\max}\}_{k \in \mathcal{K}}) \in \mathbb{R}^{K^2+K}. \quad (10)$$

Algorithm 1 Preference-Conditioned MO-DDPG for Multiuser SWIPT

1) *Initialization*

- 1: Initialize actor $\mu(\cdot|\theta)$ and vector critic $\mathbf{Q}(\cdot|\phi)$ with random weights
- 2: Initialize target networks $\theta' \leftarrow \theta$, $\phi' \leftarrow \phi$, and replay buffer $\mathcal{D}$

2) *Preference-Conditioned Training*

- 3: **for** episode = 1 $\dots$ M **do**
- 4: Sample preference vector $\omega \sim \text{Dirichlet}(\alpha)$
- 5: Reset environment and initialize battery levels $b_k(0) \sim \mathcal{U}(0, B_{\max})$
- 6: **for** $t = 0 \dots T - 1$ **do**
- 7: Select action $\mathbf{a}[t] = \mu(\mathbf{s}[t], \omega | \theta) + \mathbf{n}[t]$
- 8: Execute action $\mathbf{a}[t]$, observe reward vector $\mathbf{r}[t]$ and next state $\mathbf{s}[t+1]$
- 9: Store transition $(\mathbf{s}[t], \mathbf{a}[t], \mathbf{r}[t], \mathbf{s}[t+1], \omega)$ in $\mathcal{D}$
- 10: Sample mini-batch of transitions from $\mathcal{D}$
- 11: Compute target vector $\mathbf{y}$ according to (17)
- 12: Update critic ϕ by minimizing loss $L(\phi)$ in (16)
- 13: Update actor θ using policy gradient $\nabla_{\theta} J$ in (18)
- 14: Update target networks (θ', ϕ') with soft-update factor κ
- 15: **end for**
- 16: **end for**

2) *Action*: The actor outputs a raw action $\mathbf{a}[t] = (\hat{p}_1[t], \dots, \hat{p}_K[t], \rho_1[t], \dots, \rho_K[t])$, in which the power weights $\hat{p}_k[t]$ are produced by a softmax layer (so that $\hat{p}_k[t] \geq 0$ and $\sum_k \hat{p}_k[t] = 1$) and the PS ratios $\rho_k[t]$ are produced by a sigmoid layer (so that $\rho_k[t] \in [0, 1]$). The transmit powers are then obtained from the softmax weights as

$$p_k[t] = P_{\max} \hat{p}_k[t], \quad k \in \mathcal{K}, \quad (11)$$

so that $\sum_{k \in \mathcal{K}} p_k[t] = P_{\max}$ holds by (11) and $\rho_k[t] \in [0, 1]$ holds by the sigmoid output range.

3) *Reward*: Instead of a scalar reward, the environment returns the three-dimensional reward vector $\mathbf{r}[t] = (r_R[t], r_E[t], r_F[t])$, whose entries correspond to the per-slot utilities in (5)–(7),

$$r_R[t] = \sum_{k \in \mathcal{K}} o_k[t] R_k[t], \quad (12)$$

$$r_E[t] = \sum_{k \in \mathcal{K}} E_k[t], \quad (13)$$

$$r_F[t] = \min_{k \in \mathcal{K}} o_k[t] R_k[t]. \quad (14)$$

For a given preference ω , the scalarized utility is obtained as

$$r_{\omega}[t] = \omega^T \mathbf{r}[t] = \omega_R r_R[t] + \omega_E r_E[t] + \omega_F r_F[t]. \quad (15)$$

B. Preference-Conditioned Actor and Critic

To approximate the rate-energy-fairness tradeoff region with a single set of parameters, both the actor $\mu(\mathbf{s}, \omega | \theta)$ and vector critic $\mathbf{Q}(\mathbf{s}, \mathbf{a}, \omega | \phi) \in \mathbb{R}^3$ are conditioned on

TABLE I
SIMULATION PARAMETERS

Parameter	Value
Antennas / users N_t, K	4, 3
Transmit-power budget $P_{\max}$	1
Antenna / ID receiver noise σ_a^2, σ_c^2	0.1
Large-scale gains β	(1.0, 0.6, 0.35)
RF-to-DC conversion efficiency η	0.6
Battery capacity $B_{\max}$ / per-slot cost E_{op}	3 / 0.3
Training episodes M / slots per episode T	10^5 / 20
Discount factor γ / soft-update κ	0.95 / 5×10^{-3}
Actor / critic learning rates $\text{lr}_{\theta}, \text{lr}_{\phi}$	10^{-4} / 10^{-3}
Replay buffer / minibatch size	2×10^6 / 128
Dirichlet parameter α	0.5

preference ω [10]. During training, each episode samples $\omega \sim \text{Dirichlet}(\alpha)$ as well as single-objective preference vectors to cover Ω . The critic minimizes the per-objective temporal-difference loss

$$L(\phi) = \mathbb{E}[\|\mathbf{y} - \mathbf{Q}(\mathbf{s}, \mathbf{a}, \omega | \phi)\|^2], \quad (16)$$

$$\mathbf{y} = \mathbf{r} + \gamma \mathbf{Q}'(\mathbf{s}', \mu'(\mathbf{s}', \omega | \theta'), \omega | \phi'), \quad (17)$$

and the actor is updated via the scalarized policy gradient

$$\nabla_{\theta} J \approx \mathbb{E} \left[\nabla_{\theta} \mu(\mathbf{s}, \omega | \theta) \nabla_{\mathbf{a}} (\omega^T \mathbf{Q}(\mathbf{s}, \mathbf{a}, \omega | \phi)) \Big|_{\mathbf{a}=\mu(\mathbf{s}, \omega | \theta)} \right]. \quad (18)$$

The target networks are updated slowly as $\theta' \leftarrow \kappa \theta + (1 - \kappa) \theta'$ and $\phi' \leftarrow \kappa \phi + (1 - \kappa) \phi'$ with soft-update factor $\kappa \ll 1$. The overall training procedure of the preference-conditioned MO-DDPG framework is summarized in Algorithm 1.

C. Online Execution

During online execution, the trained actor network directly determines control actions given any desired preference ω^* . At each slot t , the BS evaluates $\mathbf{a}[t] = \mu(\mathbf{s}[t], \omega^*)$. Sweeping ω^* across the preference space Ω produces an approximation of the rate-energy-fairness tradeoff region using a single trained model. In addition, dynamically adjusting ω^* enables real-time priority reconfiguration without retraining.

IV. SIMULATION RESULTS

Simulation parameters are summarized in Table I. Initial battery levels are sampled as $b_k(0) \sim \mathcal{U}(0, B_{\max})$. We compare the proposed single MO-DDPG network against (i) six preference-specialized DDPG baselines trained separately for fixed preference vectors, and (ii) fixed sum-rate-only ($\rho_k = 1$), energy-harvesting-only ($\rho_k = 0$), and equal-split ($\rho_k = 0.5$) baseline schemes.

Fig. 2 shows the pairwise projections of the achieved sum rate, harvested energy, and user fairness. By sweeping the preference vector, the single trained MO-DDPG network produces tradeoff points comparable to those obtained by the six independently trained DDPG baselines. Although the specialized policies achieve slightly better performance at their respective target preferences, the proposed model remains within 10% of their performance across the evaluated preferences while reducing overall training and storage costs.

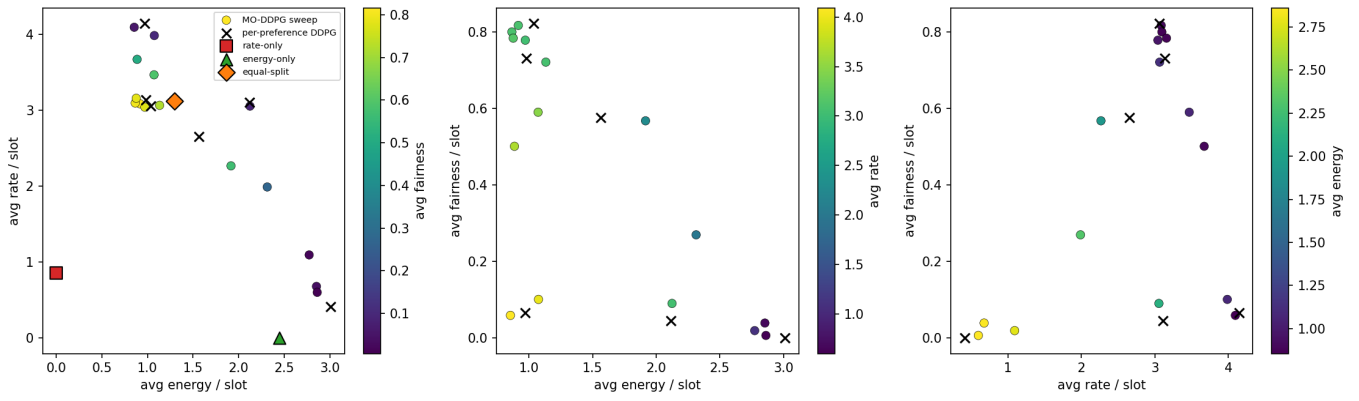


Fig. 2. Pairwise projections of the rate-energy-fairness tradeoff region, comparing the proposed single MO-DDPG network against preference-specialized DDPG baselines and fixed reference schemes.

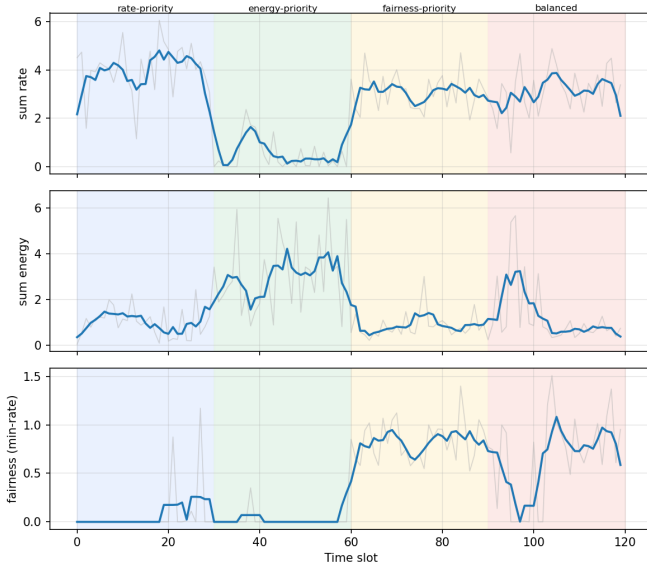


Fig. 3. Real-time preference adaptation of the proposed MO-DDPG policy under time-varying channel realizations, where operating preferences are updated every 30 slots without policy retraining.

For example, in the high-energy region, the proposed policy attains a slightly lower maximum harvested energy than the specialized DDPG baseline, while achieving a higher sum rate. This difference may reflect the compromise involved in learning a shared policy across multiple preferences rather than optimizing exclusively for a single target preference. The fixed baseline schemes serve as non-adaptive reference points.

Fig. 3 demonstrates online operation under time-varying channel conditions, where the operating preference is updated every 30 slots. The single trained policy responds to changing priorities, dynamically adjusting performance to emphasize sum rate, harvested energy, or user fairness depending on the active preference phase without requiring network retraining or online re-optimization.

V. CONCLUSION

In this paper, we formulated multiuser PS-SWIPT with finite batteries as a battery-aware multi-objective MDP over sum rate, harvested energy, and user fairness, and solved it with a preference-conditioned MO-DDPG algorithm that approximates the tradeoff region using a single policy network. Simulation results demonstrate that the single network achieves performance close to individually trained preference-specialized baseline policies while lowering training and storage requirements, and enables real-time operating priority adjustments during deployment. The proposed preference-conditioned framework thus offers a flexible and practical resource management solution for multiuser SWIPT systems.

REFERENCES

- [1] L. R. Varshney, "Transporting information and energy simultaneously," in *Proc. IEEE ISIT*, 2008, pp. 1612–1616.
- [2] R. Zhang and C. K. Ho, "MIMO broadcasting for simultaneous wireless information and power transfer," *IEEE Trans. Wireless Commun.*, vol. 12, no. 5, pp. 1989–2001, May 2013.
- [3] X. Lu, P. Wang, D. I. Kim, D. Niyato, and Z. Han, "Wireless networks with RF energy harvesting: A contemporary survey," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 2, pp. 757–789, 2nd Quart. 2015.
- [4] Q. Shi, L. Liu, W. Xu, and R. Zhang, "Joint transmit beamforming and receive power splitting for MISO SWIPT systems," *IEEE Trans. Wireless Commun.*, vol. 13, no. 6, pp. 3269–3280, Jun. 2014.
- [5] D. W. K. Ng, E. S. Lo, and R. Schober, "Multi-objective resource allocation for secure communication in cognitive radio networks with wireless information and power transfer," *IEEE Trans. Veh. Technol.*, vol. 65, no. 5, pp. 3166–3184, May 2015.
- [6] T. P. Lillicrap *et al.*, "Continuous control with deep reinforcement learning," in *Proc. ICLR*, 2016.
- [7] S. Hwang, H. Kim, H. Lee, and I. Lee, "Multi-agent deep reinforcement learning for distributed resource management in wirelessly powered communication networks," *IEEE Trans. Veh. Technol.*, vol. 69, no. 11, pp. 14055–14060, Nov. 2020.
- [8] M. Kim, H. Lee, S. Hwang, M. Debbah, and I. Lee, "Cooperative multi-agent deep reinforcement learning methods for UAV-aided mobile edge computing networks," *IEEE Internet Things J.*, vol. 11, no. 23, pp. 38040–38053, Dec. 2024.
- [9] Z. Zhu, M. Gong, M. Zhang, Q. Wu, G. Sun, Z. Chu, X. Li, and I. Lee, "Reinforcement learning based resource allocation in IRS assisted SWIPT systems," *IEEE Trans. Veh. Technol.*, vol. 74, no. 4, pp. 6790–6794, Apr. 2025.
- [10] A. Abels, D. M. Roijers, T. Lenaerts, A. Nowé, and D. Steckelmacher, "Dynamic weights in multi-objective deep reinforcement learning," in *Proc. ICML*, 2019, pp. 11–20.